SIGMOD PODS 2025 BERLIN GERMANY

Navigating Labels and Vectors: A Unified Approach to Filtered Approximate Nearest Neighbor Search

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1 Problem Statement

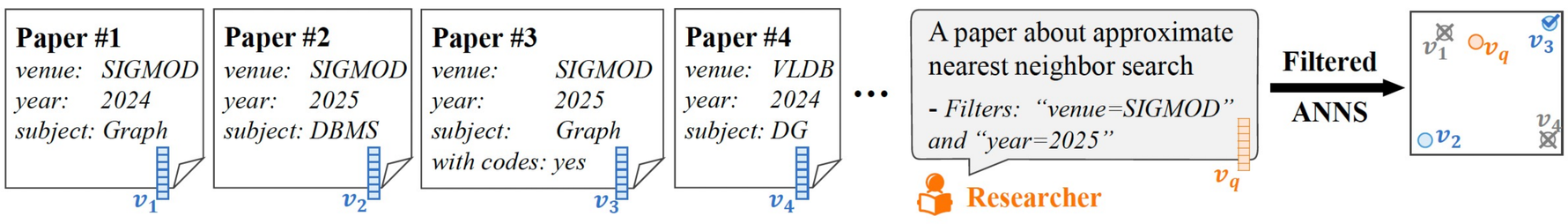
Filtered Approximate Nearest Neighbor Search

- Dataset: a set of vectors v_i, each has a label set f_i
- Query: a query vector v_q, and a query label set f_q
- Task: Approximately find top-k similar vectors v_i s.t. $f_q \subseteq f_i$

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Categorical labels with AND predicate:

```
SELECT Top-k vectors
FROM dataset
WHERE
      LabelX=1 AND
      LabelY=2 AND …
```



2 Methodology

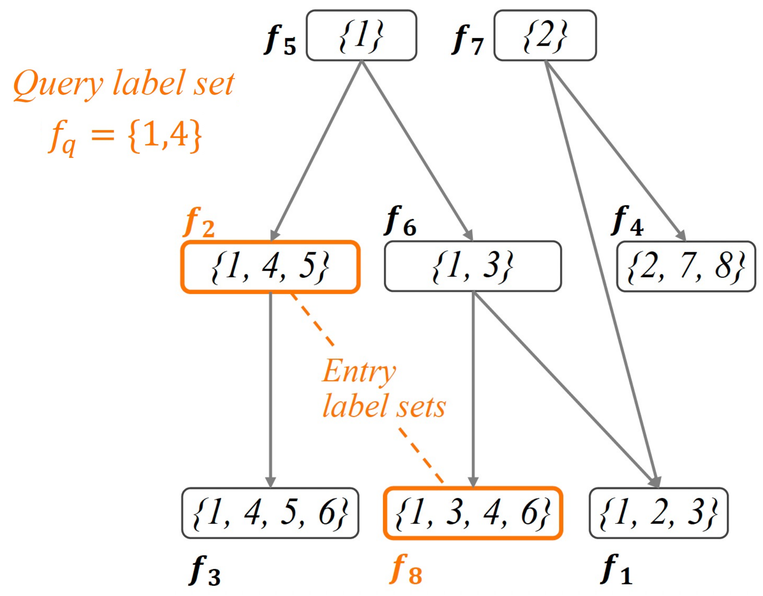
- **Versatility:** Support any size of f_q and f_i
- **Fidelity:** Only visit the filtered vectors
- **Completeness:** Return enough answers
- **Adaptability:** Support different graph ANNS indices

Key Observation

- Task: Approximately find top-k similar vectors v_i s.t. $f_q \subseteq f_i$
- If $f_i \subseteq f_j$, when f_i satisfies the filter condition, f_j also satisfies ($f_q \subseteq f_i \subseteq f_j$)

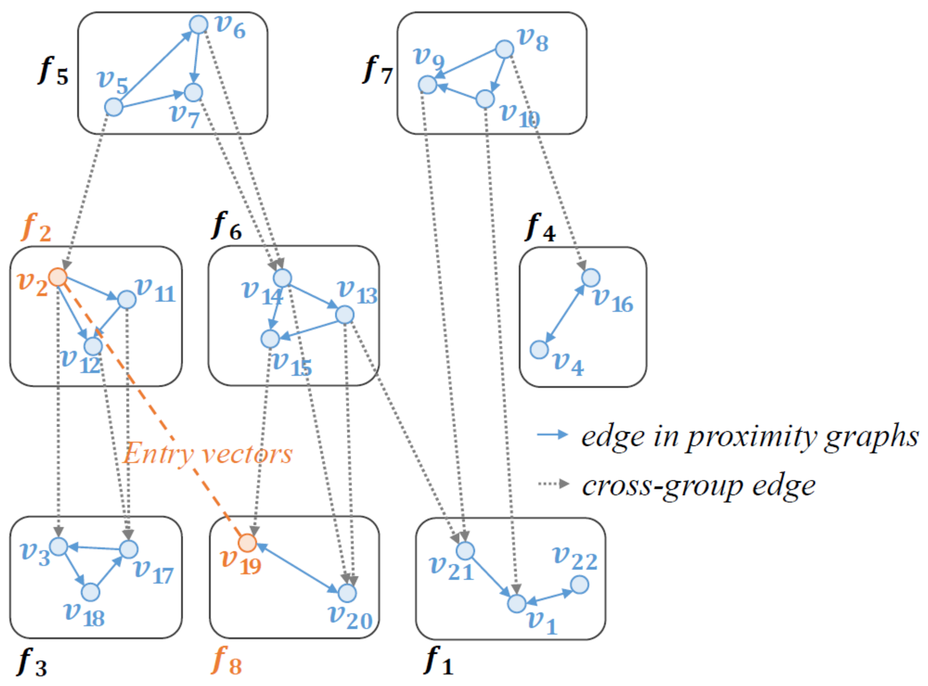
Label Navigating Graph

- Build the containment relations for all f_i
- Find some entry label sets



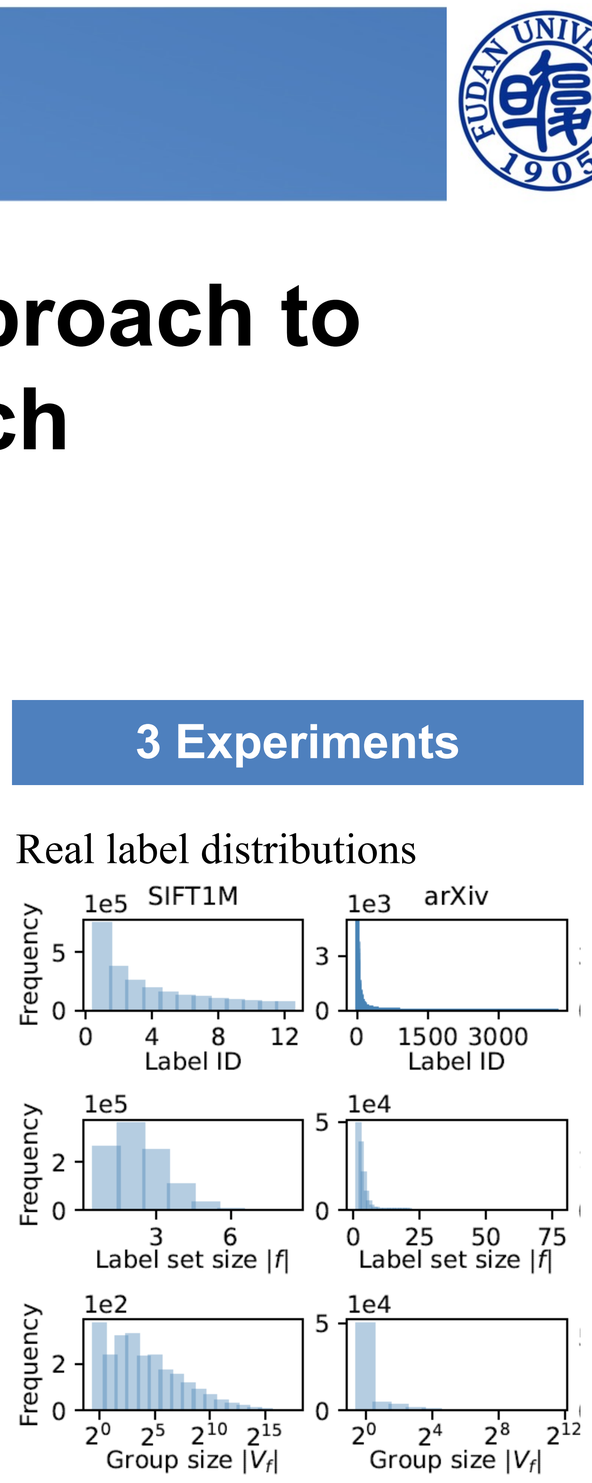
Unified Navigating Graph (UNG)

- Build proximity graph for each group with same f_i
- Add directional cross-group edges

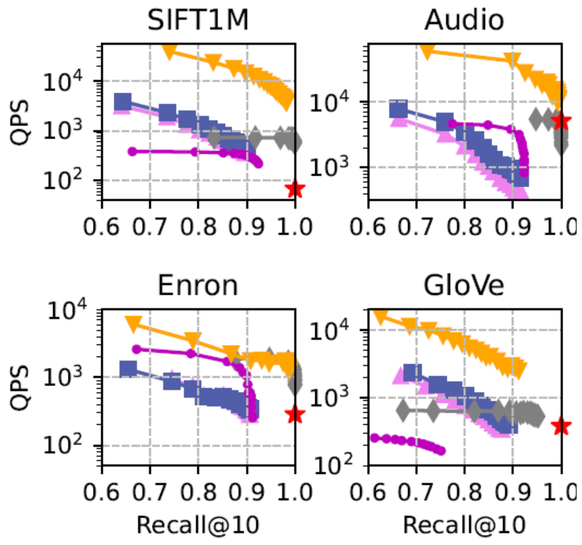


3 Experiments

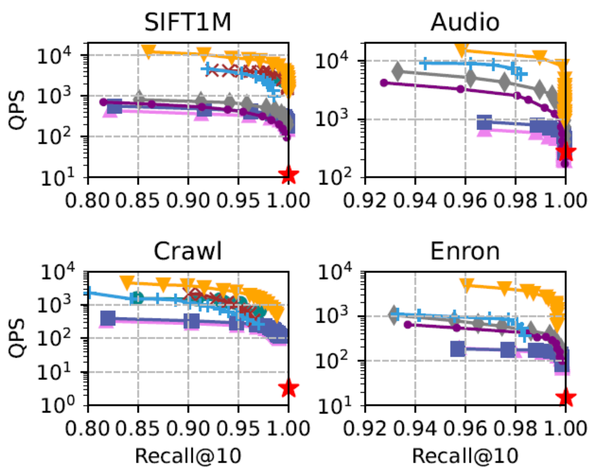
Real label distributions



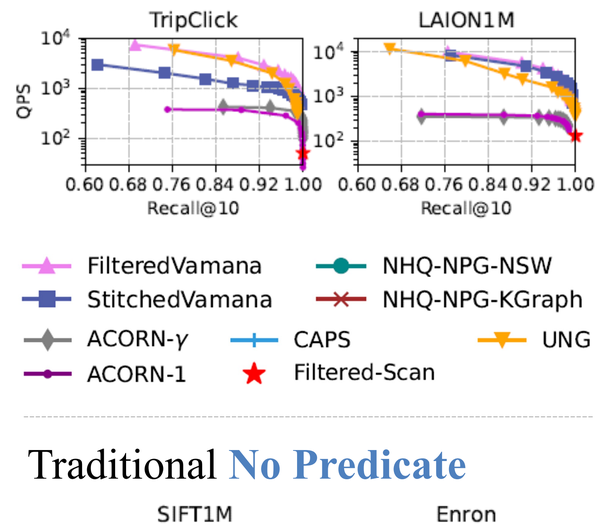
Predicate **AND**



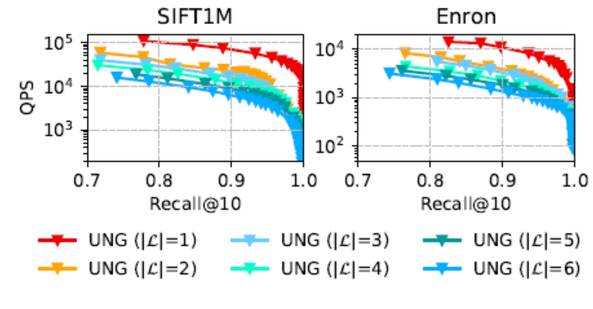
Predicate **EQUAL**



Predicate **OR**



Traditional **No Predicate**



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